Forces on two objects

antocciphysics | TEACHER KEY | CPA

Learning target: Identify an equal-and-opposite interaction pair on different objects.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 11; 3 course days later.*
6 m/s²; the mass doubles at the same net force. Keep the net force fixed, including resistance.

02. Explain a worked example (worked)
Equal opposite interaction arrows on different bodies; acceleration also depends on mass.

03. Name both forces (model)
Hand on wall; wall on hand. Equal forces on different objects.

04. Heavy and light carts (practice)
Equal forces; lighter cart greater acceleration for same force.

05. Balanced or a pair? (reason)
No; both act on book. Actual partners are book on Earth and book on table.

06. Choose a method (mixed) *Review: Unit 1, Day 2; 12 course days later.*
Distance 18 m; displacement +2 m (right); final position 8 m.

07. Exit check (exit)
Ground pushes foot forward; acts on foot. Forward external force can accelerate person.

08. STRETCH: Apply it another way (stretch)
Any inverse mass ratio, e.g. 20 and 60 kg; lighter skater accelerates three times as much.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 11; 3 course days later.*
Net force 0.24 N; new a≈0.3 m/s². Extra resistance reduces net force and acceleration if driving force is unchanged.

02. Explain a worked example (worked)
Equal opposite interaction arrows on different bodies; acceleration also depends on mass.

03. Equal forces, different results (model)
aA 12 m/s²; aB 3 m/s²; opposite directions.

04. Identify actual partners (practice)
Book-on-Earth and book-on-table. Weight and normal both act on book; pair forces act on different objects.

05. Horse plus cart (reason)
Internal horse cart forces cancel in total; external ground force on horse can exceed resistance. Need net external force.

06. Choose a method (mixed) *Review: Unit 1, Day 2; 12 course days later.*
Distance 21 m; displacement -3 m; final position +4 m.

07. Exit check (exit)
Equal opposite forces; smaller-mass craft larger a. Comparison assumes other forces negligible or compares contribution from this interaction only.

08. STRETCH: Apply it another way (stretch)
System acceleration 10/5=2 m/s². Contact on 3 kg cart is 6 N; reaction on 2 kg cart is 6 N opposite the applied 10 N.